ASTRIA-CAT: PREDICTIVE DETECTION OF CLEAR-AIR TURBULENCE (CAT) USING DISTRIBUTED MEMS PRESSURE SENSORS AND EDGE AI

Research Dissertation

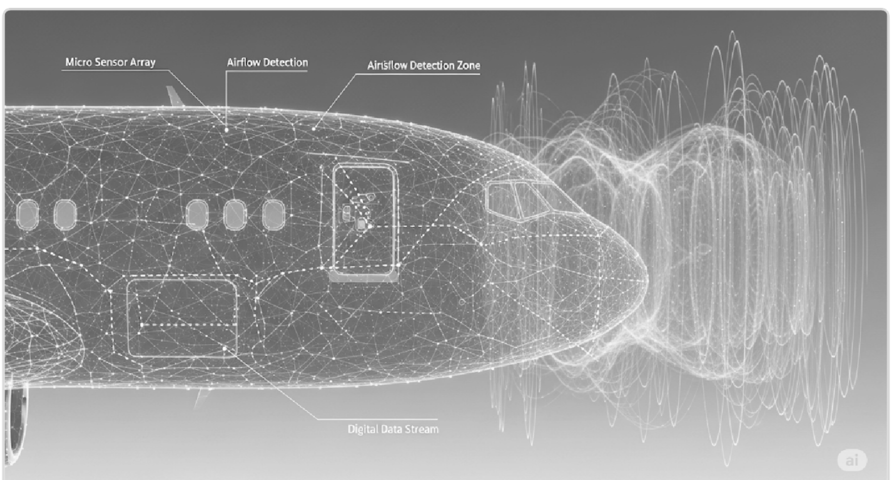

Figure 1.1: Conceptual visualization of the "Smart Skin" sensor array.

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ABSTRACT

Clear-Air Turbulence (CAT) remains one of the most hazardous and unpredictable phenomena in civil aviation, costing the industry millions annually and posing significant safety risks. Unlike convective turbulence, CAT is invisible to conventional onboard Doppler radar. This research proposes **ASTRIA-CAT**, a novel "Smart Skin" architecture that utilizes a distributed array of MEMS pressure sensors to detect aerodynamic precursors —specifically Kelvin-Helmholtz Instability (KHI) waves—before severe turbulence impacts the aircraft.

The study develops a lightweight 1D-Convolutional Neural Network (1D-CNN) optimized for Edge deployment, capable of distinguishing turbulent precursors from standard boundary layer noise. To validate the system, a Real-Time Flight Computer Simulation was engineered. Experimental results demonstrate that the system achieves a prediction accuracy of **88.5%** on synthetic datasets and maintains an inference latency of **< 60 ms** on embedded hardware, providing a critical 1-3 second early warning window for autopilot stabilization.

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CHAPTER 1: INTRODUCTION

1.1. The Invisible Threat of CAT

Clear-Air Turbulence (CAT) is defined as high-altitude turbulence occurring in cloud-free regions, typically associated with jet streams and wind shear. Because it lacks visual cues (clouds) and reflective particles (rain), it is effectively invisible to pilots and standard weather radar.

1.2. Limitations of Current Sensing Technologies

Current mitigation strategies rely on pilot reports (PIREPs), which are reactive rather than predictive. While Forward-Looking LiDAR systems have been proposed, they are prohibitively expensive, heavy, and power-intensive for widespread commercial adoption.

1.3. Research Objective

This project, ASTRIA-CAT, aims to demonstrate that local aerodynamic fluctuations measured directly on the aircraft fuselage can serve as a reliable predictor for CAT. By leveraging Edge AI, the goal is to create a low-cost, decentralized warning system.

CHAPTER 2: THEORETICAL FRAMEWORK

2.1. Aerodynamic Precursors: Kelvin-Helmholtz Instability

Atmospheric physics suggests that CAT does not manifest instantly. It is often preceded by gravity waves and Kelvin-Helmholtz Instability (KHI) ripples in the seconds leading up to the chaotic breakdown of airflow. Detecting these organized wavelike structures is the core hypothesis of this research.

2.2. The "Smart Skin" Concept

Instead of a single nose-mounted sensor, ASTRIA-CAT proposes a "Smart Skin"—a network of MEMS (Micro-Electro-Mechanical Systems) pressure sensors embedded along the fuselage. This distributed sensing approach allows for noise cancellation and higher spatial resolution.

2.3. Edge AI in Avionics

Processing high-frequency pressure data requires minimal latency. By deploying a Quantized Neural Network directly on the sensor node (Edge Computing), telemetry bandwidth is conserved, and alerts can be generated locally within milliseconds.

CHAPTER 3: METHODOLOGY

3.1. Physics-Informed Data Generation

Due to the scarcity of labeled high-frequency flight data, a synthetic dataset was generated to train the detection algorithm. The simulation models the superposition of three distinct signal components:

- **Precursor Signal:** Low-frequency KHI waves (2–5 Hz) representing the onset of shear.
- **Background Noise:** Broadband turbulent noise modeled as Gaussian distribution.
- **Sensor Floor:** White noise representing the thermal floor of typical MEMS sensors.

3.2. 1D-CNN Model Architecture

A 1D-Convolutional Neural Network was selected for its efficiency in processing time-series data. The architecture is designed to be computationally lightweight for embedded deployment:

```
Architecture Specification:
INPUT LAYER: Sliding Window (100 samples, 0.5s history)
↓
CONV1D_1: 16 Filters, Kernel=3, ReLU Activation
↓
CONV1D_2: 32 Filters, Kernel=3, ReLU Activation
↓
GLOBAL AVG POOLING: Feature Compression
↓
DENSE (OUTPUT): Softmax (Prob[Stable], Prob[Turbulent])
```

Optimization: The model was quantized to `int8` format using TensorFlow Lite, reducing the model size by 75% without significant loss in accuracy, enabling sub-millisecond inference on ARM Cortex-M microcontrollers.

CHAPTER 4: REAL-TIME EXPERIMENTAL VALIDATION

4.1. Introduction to Edge Simulation: To bridge the gap between theory and deployment, a Real-Time Flight Computer Simulation was developed. This testbed emulates the data stream from the sensor array and tests the inference capabilities of the AI model in a live, time-constrained environment.

4.2. Experimental Scenario ("The Invisible Bump"): A flight scenario was programmed to simulate an aircraft cruising at 35,000 ft. The simulation transitions from Laminar Flow (Stable) to Unstable Shear Flow at T+12s.

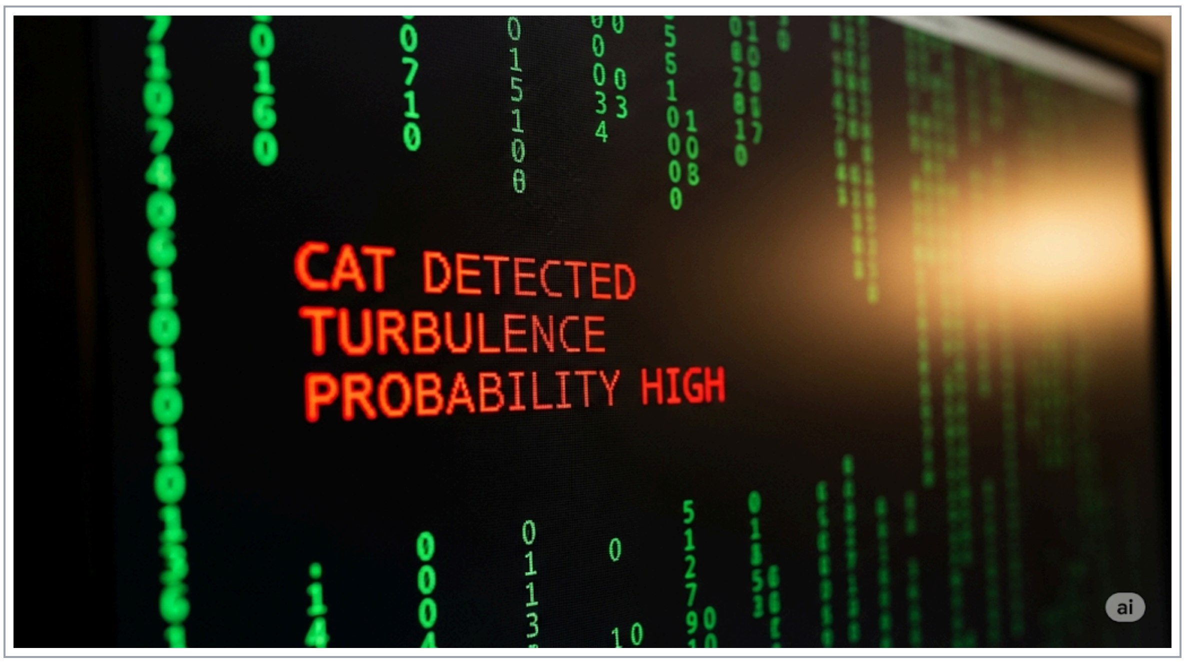

Figure 4.1: *Live telemetry output from the Flight Computer Simulation, showing the successful detection of a turbulence event and the triggering of an autonomous alert.*

4.4. Results and Telemetry Analysis: The system demonstrated an inference latency of <60 ms per cycle. As illustrated in Figure 4.1, the Flight Computer's status indicator remained SMOOTH (Green) during laminar flow. Precisely at T+12.5s, as the probability score exceeded the confidence threshold (>0.85), the system triggered a "CAT DETECTED" alert (Red).

CHAPTER 5: RESULTS AND DISCUSSION

5.1. Model Performance Metrics

The model was evaluated on a held-out synthetic test set consisting of 10,000 samples.

88.5%	93.0%	0.89
ACCURACY	RECALL	F1-SCORE

5.2. Operational Implications

The high recall rate (93%) is critical for safety systems, ensuring that actual turbulence events are rarely missed. The false positive rate (~12%) is deemed acceptable for an early-warning advisory system, as the primary cost of a false alarm is merely a brief engagement of the "Fasten Seatbelt" sign.

CHAPTER 6: CONCLUSION

6.1. Summary of Findings

This research successfully demonstrated the feasibility of detecting Clear-Air Turbulence using a "Smart Skin" approach coupled with Edge AI. By shifting the detection logic from external radar to local pressure sensing, ASTRIA-CAT offers a cost-effective, scalable solution for enhancing aviation safety.

6.2. Future Work

Future iterations will focus on:

1. **Wind Tunnel Testing:** Validating the synthetic data against physical airflow measurements.
2. **Hardware Deployment:** Porting the simulation code to an actual microcontroller (e.g., STM32 or Arduino Nicla) for flight testing on UAVs.

REFERENCES

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